

UNIT III – SQL RELATIONSHIPS

1. SQL Relationship

In a relational database, data is usually stored in **multiple tables**. A relationship is used to connect the data in one table with the data in another table.

For example, consider two tables:

Users

id	name
1	John
2	May

Orders

Id	User_id	product
101	1	Laptop
102	2	Mobile
103	3	Mobile

Here,

Users.id and Orders.user_id are related.

So:

Users.id → Orders.user_id

This relationship allows us to find which user placed which order.

2. Types of SQL Relationships

There are **three important types** of relationships:



4. JOIN

JOIN is used to combine data from different tables.

Important JOIN types:

LEFT JOIN

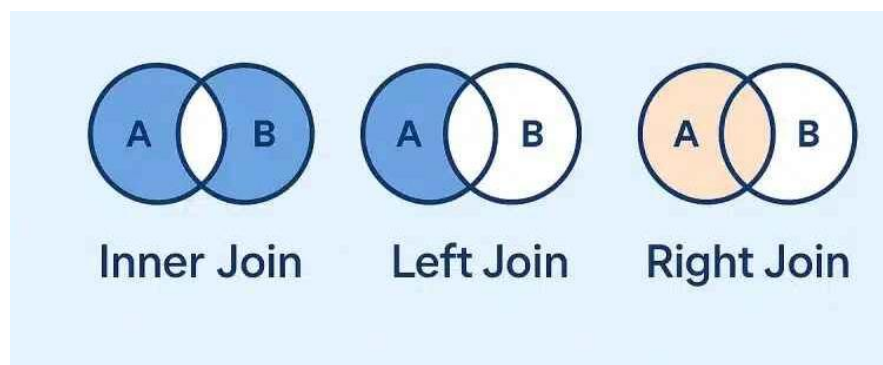
- Returns all records from the **left table**.
- If there is no matching record, it gives NULL.

RIGHT JOIN

- Returns all records from the **right table**.
- Missing matching data becomes NULL.

INNER JOIN

- Returns **only matching records** from both tables.



5. ON Keyword

ON specifies **how two tables are connected**.

Example:

```
ON users.id = orders.user_id
```

This is very important when writing JOIN queries.

6. Duplicate Column Names

If both tables have the same column name like id, specify the table name:

```
SELECT orders.id
```

instead of:

```
SELECT id
```

7. Subquery

A **subquery** is a SQL query written inside another SQL query.

It is useful when the result of one query is needed as a condition for another query.

Example:

```
SELECT *  
FROM orders  
WHERE user_id = (SELECT id FROM users WHERE name = 'John');
```

8. = vs IN in Subquery

=

- Use when the subquery returns **one value**.

IN

- Use when the subquery can return **multiple values**.

Example:

```
WHERE user_id = (subquery)
```

```
WHERE user_id IN (subquery)
```

Movies, OTT & Entertainment – SQL Major Project Dataset

This dataset is **student-friendly**, **industry-inspired**, and suitable for a **major SQL project**.

It supports **single-table** → **multi-table learning**, analytics, and reporting.

1. MOVIES TABLE

Table Structure

```
CREATE TABLE movies (  
    movie_id INT PRIMARY KEY,  
    movie_name VARCHAR(100),  
    genre VARCHAR(30),  
    language VARCHAR(30),  
    release_year INT,  
    duration INT, -- minutes  
    rating DECIMAL(3,1)  
);
```

Insert Data

```
INSERT INTO movies VALUES  
(1, 'Jai Bhim', 'Drama', 'Tamil', 2021, 164, 8.8),  
(2, 'RRR', 'Action', 'Telugu', 2022, 187, 8.6),  
(3, 'KGF Chapter 2', 'Action', 'Kannada', 2022, 168, 8.4),  
(4, 'Vikram', 'Action', 'Tamil', 2022, 175, 8.3),  
(5, 'Soorarai Pottru', 'Drama', 'Tamil', 2020, 153, 8.7),  
(6, 'Drishyam 2', 'Thriller', 'Malayalam', 2021, 152, 8.2),  
(7, 'Dangal', 'Sports', 'Hindi', 2016, 161, 8.4),  
(8, 'Baahubali 2', 'Fantasy', 'Telugu', 2017, 171, 8.2),  
(9, 'Andhadhun', 'Thriller', 'Hindi', 2018, 139, 8.2),  
(10, 'Kantara', 'Drama', 'Kannada', 2022, 148, 8.3),  
(11, 'Leo', 'Action', 'Tamil', 2023, 164, 7.8),  
(12, 'Jawan', 'Action', 'Hindi', 2023, 169, 7.9),  
(13, 'Pushpa', 'Action', 'Telugu', 2021, 179, 7.6),  
(14, '3 Idiots', 'Comedy', 'Hindi', 2009, 170, 8.4),  
(15, 'Premam', 'Romance', 'Malayalam', 2015, 156, 8.3);
```

2. ACTORS TABLE

```
CREATE TABLE actors (  
    actor_id INT PRIMARY KEY,  
    actor_name VARCHAR(50),  
    gender VARCHAR(10)  
);
```

```
INSERT INTO actors VALUES  
(1, 'Surya', 'Male'),  
(2, 'Vijay', 'Male'),  
(3, 'Kamal Haasan', 'Male'),  
(4, 'Yash', 'Male'),  
(5, 'Ram Charan', 'Male'),  
(6, 'Jr NTR', 'Male'),  
(7, 'Aamir Khan', 'Male'),  
(8, 'Prithviraj', 'Male'),  
(9, 'Rajkummar Rao', 'Male'),  
(10, 'Alia Bhatt', 'Female');
```

3. MOVIE_CAST TABLE (Many-to-Many)

```
CREATE TABLE movie_cast (  
    movie_id INT,  
    actor_id INT,  
    role_name VARCHAR(50),  
    PRIMARY KEY (movie_id, actor_id)  
);
```

```
INSERT INTO movie_cast VALUES  
(1,1, 'Lawyer'),  
(2,5, 'Freedom Fighter'),  
(2,6, 'Revolutionary'),  
(3,4, 'Rocky'),  
(4,3, 'Agent Vikram'),  
(5,1, 'Pilot'),  
(6,8, 'Father'),  
(7,7, 'Wrestler'),  
(8,5, 'Prince'),  
(9,9, 'Blind Pianist'),  
(10,4, 'Tribal Hero'),  
(11,2, 'Gangster'),  
(12,2, 'Vigilante'),  
(13,4, 'Smuggler'),  
(14,7, 'Student'),  
(15,8, 'College Student');
```

4. OTT PLATFORMS TABLE

```
CREATE TABLE ott_platforms (  
    platform_id INT PRIMARY KEY,  
    platform_name VARCHAR(30),  
    launch_year INT  
);
```

```
INSERT INTO ott_platforms VALUES  
(1, 'Netflix', 2016),  
(2, 'Amazon Prime', 2016),  
(3, 'Disney+ Hotstar', 2019),  
(4, 'Zee5', 2018),  
(5, 'Sony LIV', 2020);
```

5. MOVIE_STREAMING TABLE

```
CREATE TABLE movie_streaming (  
    movie_id INT,  
    platform_id INT,  
    streaming_year INT,  
    PRIMARY KEY (movie_id, platform_id)  
);
```

```
INSERT INTO movie_streaming VALUES  
(1, 2, 2021),  
(2, 1, 2022),  
(3, 2, 2022),  
(4, 3, 2022),  
(5, 2, 2020),  
(6, 3, 2021),  
(7, 1, 2017),  
(8, 1, 2018),  
(9, 2, 2019),  
(10, 4, 2022),  
(11, 3, 2023),  
(12, 1, 2023),  
(13, 2, 2021),  
(14, 1, 2010),  
(15, 2, 2016);
```

6. USERS TABLE

```
CREATE TABLE users (  
    user_id INT PRIMARY KEY,  
    user_name VARCHAR(50),  
    city VARCHAR(30)  
);
```

```
INSERT INTO users VALUES  
(1, 'Arun', 'Chennai'),  
(2, 'Divya', 'Coimbatore'),
```


5. Display movies with rating above 8.5.

```
SELECT * FROM movies WHERE rating > 8.5;
```

	movie_id	movie_name	genre	language	release_year	duration	rating
▶	1	Jai Bhim	Drama	Tamil	2021	164	8.8
	2	RRR	Action	Telugu	2022	187	8.6
	5	Soorarai Pottru	Drama	Tamil	2020	153	8.7

6. Show distinct movie languages.

```
SELECT DISTINCT language FROM movies;
```

	language
▶	Tamil
	Telugu
	Kannada
	Malayalam
	Hindi

7. Display top 5 highest-rated movies.

```
SELECT * FROM movies ORDER BY rating DESC LIMIT 5;
```

	movie_id	movie_name	genre	language	release_year	duration	rating
▶	1	Jai Bhim	Drama	Tamil	2021	164	8.8
	5	Soorarai Pottru	Drama	Tamil	2020	153	8.7
	2	RRR	Action	Telugu	2022	187	8.6
	3	KGF Chapter 2	Action	Kannada	2022	168	8.4
	7	Dangal	Sports	Hindi	2016	161	8.4

8. Count total number of movies.

```
SELECT COUNT(*) FROM movies;
```

	total_movies
▶	15

9. Find average movie duration.

```
SELECT AVG(duration) FROM movies;
```

	average_duration
▶	163.7333

10. Display movies ordered by release year.

```
SELECT * FROM movies ORDER BY release_year;
```

	movie_id	movie_name	genre	language	release_year	duration	rating
▶	14	3 Idiots	Comedy	Hindi	2009	170	8.4
	15	Premam	Romance	Malayalam	2015	156	8.3
	7	Dangal	Sports	Hindi	2016	161	8.4
	8	Baahubali 2	Fantasy	Telugu	2017	171	8.2
	9	Andhadhun	Thriller	Hindi	2018	139	8.2
	5	Soorai Pottu	Drama	Tamil	2020	153	8.7
	1	Jai Bhim	Drama	Tamil	2021	164	8.8
	6	Drishyam 2	Thriller	Malayalam	2021	152	8.2
	13	Pushpa	Action	Telugu	2021	179	7.6
	2	RRR	Action	Telugu	2022	187	8.6
	3	KGF Chapter 2	Action	Kannada	2022	168	8.4
	4	Vikram	Action	Tamil	2022	175	8.3
	10	Kantara	Drama	Kannada	2022	148	8.3
	11	Leo	Action	Tamil	2023	164	7.8
	12	Jawan	Action	Hindi	2023	169	7.9

PART B – GROUP BY & Aggregates

11. Genre-wise movie count.

```
SELECT genre, COUNT(*) FROM movies GROUP BY genre;
```

	genre	movie_count
▶	Drama	3
	Action	6
	Thriller	2
	Sports	1
	Fantasy	1
	Comedy	1
	Romance	1

12. Language-wise average rating.

```
SELECT language, AVG(rating) FROM movies GROUP BY language;
```

	language	average_rating
▶	Tamil	8.40000
	Telugu	8.13333
	Kannada	8.35000
	Malayalam	8.25000
	Hindi	8.22500

13. Find maximum rating per genre.

```
SELECT genre, MAX(rating) FROM movies GROUP BY genre;
```

	genre	maximum_rating
►	Drama	8.8
	Action	8.6
	Thriller	8.2
	Sports	8.4
	Fantasy	8.2
	Comedy	8.4
	Romance	8.3

14.Count movies released each year.

```
SELECT release_year, COUNT(*) FROM movies GROUP BY release_year;
```

	release_year	movis_count_per_year
►	2021	3
	2022	4
	2020	1
	2016	1
	2017	1
	2018	1
	2023	2
	2009	1
	2015	1

15.Display genres having more than 2 movies.

```
SELECT genre, COUNT(*) FROM movies GROUP BY genre HAVING COUNT(*) > 2;
```

	genre	movies_count
►	Drama	3
	Action	6

PART C – JOIN Queries (Movies, OTT, Streaming)

16. List movies with their OTT platforms.

```
SELECT m.movie_name, o.platform_name  
FROM movies m  
JOIN movie_streaming ms ON m.movie_id = ms.movie_id  
JOIN ott_platforms o ON ms.platform_id = o.platform_id;
```

	movie_name	platform_name
▶	Jai Bhim	Amazon Prime
	RRR	Netflix
	KGF Chapter 2	Amazon Prime
	Vikram	Disney+ Hotstar
	Soorai Pottu	Amazon Prime
	Drishyam 2	Disney+ Hotstar
	Dangal	Netflix
	Baahubali 2	Netflix
	Andhadhun	Amazon Prime
	Kantara	Zee5
	Leo	Disney+ Hotstar
	Jawan	Netflix
	Pushpa	Amazon Prime
	3 Idiots	Netflix
	Premam	Amazon Prime

17.Display movies available on Netflix.

```
SELECT m.movie_name
FROM movies m
JOIN movie_streaming ms ON m.movie_id = ms.movie_id
JOIN ott_platforms o ON ms.platform_id = o.platform_id
WHERE o.platform_name = 'Netflix';
```

	movie_name
▶	RRR
	Dangal
	Baahubali 2
	Jawan
	3 Idiots

18.Count movies per OTT platform.

```
SELECT o.platform_name, COUNT(*)
FROM ott_platforms o
JOIN movie_streaming ms ON o.platform_id = ms.platform_id
GROUP BY o.platform_name;
```

	platform_name	COUNT(*)
▶	Amazon Prime	6
	Netflix	5
	Disney+ Hotstar	3
	Zee5	1

19. Find OTT platform with maximum movies.

```
SELECT o.platform_name, COUNT(*) AS total
FROM ott_platforms o
JOIN movie_streaming ms ON o.platform_id = ms.platform_id
GROUP BY o.platform_name
ORDER BY total DESC
LIMIT 1;
```

	platform_name	total
▶	Amazon Prime	6

20. Display movies streamed after 2021.

```
SELECT m.movie_name
FROM movies m
JOIN movie_streaming ms ON m.movie_id = ms.movie_id
WHERE ms.streaming_year > 2021;
```

	movie_name
▶	RRR
	KGF Chapter 2
	Vikram
	Kantara
	Leo
	Jawan

PART D – Actor & Cast Analysis

21. List actors and their movies.

```
SELECT a.actor_name, m.movie_name
FROM actors a
JOIN movie_cast mc ON a.actor_id = mc.actor_id
JOIN movies m ON mc.movie_id = m.movie_id;
```

	actor_name	movie_name
▶	Surya	Jai Bhim
	Ram Charan	RRR
	Jr NTR	RRR
	Yash	KGF Chapter 2
	Kamal Haasan	Vikram
	Surya	Soorai Pottu
	Prithviraj	Drishyam 2
	Aamir Khan	Dangal
	Ram Charan	Baahubali 2
	Rajkummar Rao	Andhadhun
	Yash	Kantara
	Vijay	Leo
	Vijay	Jawan
	Yash	Pushpa

22.Count number of movies per actor.

```
SELECT a.actor_name, COUNT(*)
FROM actors a
JOIN movie_cast mc ON a.actor_id = mc.actor_id
GROUP BY a.actor_name;
```

	actor_name	COUNT(*)
▶	Surya	2
	Ram Charan	2
	Jr NTR	1
	Yash	3
	Kamal Haasan	1
	Prithviraj	2
	Aamir Khan	2
	Rajkummar Rao	1
	Vijay	2

23. Actor who acted in maximum movies.

```
SELECT a.actor_name, COUNT(*) AS total
FROM actors a
JOIN movie_cast mc ON a.actor_id = mc.actor_id
GROUP BY a.actor_name
ORDER BY total DESC
LIMIT 1;
```

	actor_name	total
▶	Yash	3

24. List movies featuring actor 'Yash'.

```
SELECT m.movie_name
FROM movies m
JOIN movie_cast mc ON m.movie_id = mc.movie_id
JOIN actors a ON mc.actor_id = a.actor_id
WHERE a.actor_name = 'Yash';
```

	movie_name
▶	KGF Chapter 2
	Kantara
	Pushpa

25. Find actors involved in action movies.

```
SELECT DISTINCT a.actor_name
FROM actors a
JOIN movie_cast mc ON a.actor_id = mc.actor_id
JOIN movies m ON mc.movie_id = m.movie_id
WHERE m.genre = 'Action';
```

	actor_name
▶	Ram Charan
	Jr NTR
	Yash
	Kamal Haasan
	Vijay

PART E – User Viewing Analysis

26. List users and movies they watched.

```
SELECT u.user_name, m.movie_name
FROM users u
JOIN viewing_history v ON u.user_id = v.user_id
JOIN movies m ON v.movie_id = m.movie_id;
```

	user_name	movie_name
▶	Arun	Jai Bhim
	Arun	Vikram
	Divya	RRR
	Rahul	KGF Chapter 2
	Sneha	Soorai Pottu
	Karthik	Dangal
	Divya	Kantara
	Rahul	Leo

27.Count movies watched by each user.

```
SELECT u.user_name, COUNT(*)
FROM users u
JOIN viewing_history v ON u.user_id = v.user_id
GROUP BY u.user_name;
```

	user_name	COUNT(*)
▶	Arun	2
	Divya	2
	Rahul	2
	Sneha	1
	Karthik	1

28.User who watched maximum movies.

```
SELECT u.user_name, COUNT(*) AS total
FROM users u
JOIN viewing_history v ON u.user_id = v.user_id
GROUP BY u.user_name
ORDER BY total DESC
LIMIT 1;
```

	user_name	total
▶	Arun	2

29. Most watched movie.

```
SELECT m.movie_name, COUNT(*) AS views
FROM movies m
JOIN viewing_history v ON m.movie_id = v.movie_id
GROUP BY m.movie_name
ORDER BY views DESC
LIMIT 1;
```

	movie_name	views
►	Jai Bhim	1

30. Total watch hours per movie.

```
SELECT m.movie_name, SUM(v.watch_hours)
FROM movies m
JOIN viewing_history v ON m.movie_id = v.movie_id
GROUP BY m.movie_name;
```

	movie_name	SUM(v.watch_hours)
►	Jai Bhim	2.5
	Vikram	3.0
	RRR	2.8
	KGF Chapter 2	3.1
	Soorarai Pottru	2.6
	Dangal	2.7
	Kantara	2.4
	Leo	2.9

PART F – Subqueries & Analytics

31. Movies with rating above average.

```
SELECT * FROM movies
WHERE rating > (SELECT AVG(rating) FROM movies);
```

	movie_id	movie_name	genre	language	release_year	duration	rating
▶	1	Jai Bhim	Drama	Tamil	2021	164	8.8
	2	RRR	Action	Telugu	2022	187	8.6
	3	KGF Chapter 2	Action	Kannada	2022	168	8.4
	4	Vikram	Action	Tamil	2022	175	8.3
	5	Soorarai Pottru	Drama	Tamil	2020	153	8.7
	7	Dangal	Sports	Hindi	2016	161	8.4
	10	Kantara	Drama	Kannada	2022	148	8.3
	14	3 Idiots	Comedy	Hindi	2009	170	8.4
	15	Premam	Romance	Malayalam	2015	156	8.3

32. Movies longer than average duration.

```
SELECT * FROM movies
WHERE duration > (SELECT AVG(duration) FROM movies);
```

	movie_id	movie_name	genre	language	release_year	duration	rating
▶	1	Jai Bhim	Drama	Tamil	2021	164	8.8
	2	RRR	Action	Telugu	2022	187	8.6
	3	KGF Chapter 2	Action	Kannada	2022	168	8.4
	4	Vikram	Action	Tamil	2022	175	8.3
	8	Baahubali 2	Fantasy	Telugu	2017	171	8.2
	11	Leo	Action	Tamil	2023	164	7.8
	12	Jawan	Action	Hindi	2023	169	7.9
	13	Pushpa	Action	Telugu	2021	179	7.6
	14	3 Idiots	Comedy	Hindi	2009	170	8.4

33. OTT platforms hosting action movies.

```
SELECT DISTINCT o.platform_name
FROM ott_platforms o
JOIN movie_streaming ms ON o.platform_id = ms.platform_id
JOIN movies m ON ms.movie_id = m.movie_id
WHERE m.genre = 'Action';
```

	platform_name
▶	Netflix
	Amazon Prime
	Disney+ Hotstar

34.Users who watched movies rated above 8.5.

```
SELECT DISTINCT u.user_name
FROM users u
JOIN viewing_history v ON u.user_id = v.user_id
JOIN movies m ON v.movie_id = m.movie_id
WHERE m.rating > 8.5;
```

	user_name
▶	Arun
	Divya
	Sneha

35.Movies not yet watched by any user.

```
SELECT movie_name FROM movies
WHERE movie_id NOT IN (SELECT movie_id FROM viewing_history);
```

	movie_name
▶	Drishyam 2
	Baahubali 2
	Andhadhun
	Jawan
	Pushpa
	3 Idiots
	Premam

PART G – Date & Reporting Queries

36.Movies watched in 2023.

```
SELECT DISTINCT m.movie_name
FROM movies m
JOIN viewing_history v ON m.movie_id = v.movie_id
WHERE YEAR(v.watch_date) = 2023;
```

	movie_name
▶	Jai Bhim
	Vikram
	RRR
	KGF Chapter 2
	Soorai Pottru
	Dangal
	Kantara
	Leo

37.Users who watched movies in multiple months.

```
SELECT u.user_name
FROM users u
JOIN viewing_history v ON u.user_id = v.user_id
GROUP BY u.user_name
HAVING COUNT(DISTINCT MONTH(v.watch_date)) > 1;
```

	user_name
▶	Arun
	Divya
	Rahul

38.Movies streamed before release year.

```
SELECT m.movie_name
FROM movies m
JOIN movie_streaming ms ON m.movie_id = ms.movie_id
WHERE ms.streaming_year < m.release_year;
```

	movie_name
--	------------

39. Latest released movie.

```
SELECT * FROM movies ORDER BY release_year DESC LIMIT 1;
```

	movie_id	movie_name	genre	language	release_year	duration	rating
▶	11	Leo	Action	Tamil	2023	164	7.8

40. Oldest movie in dataset.

```
SELECT * FROM movies ORDER BY release_year LIMIT 1;
```

	movie_id	movie_name	genre	language	release_year	duration	rating
▶	14	3 Idiots	Comedy	Hindi	2009	170	8.4